AI in Economics

The U.S. Sports Analytics Industry: AI, Growth, and Data Potential

Diego Hernandez Romero
Independent Researcher
December 21, 2025

Abstract

This paper explores the rapid growth and evolution of the U.S. sports analytics industry through the lens of artificial intelligence (AI), as studied in the Res-Econ 490 AI course. Drawing on industry reports, government data, investment trends, and technology adoption surveys, the paper analyzes how AI is transforming performance management, decision-making, and fan engagement in sports. It identifies key indicators of industry growth and proposes further research using publicly available datasets from the BLS, BEA, Crunchbase, and NSF to support economic planning and personal career development.

A. Introduction

Artificial intelligence is transforming the structure of modern industries — not only through automation and cost reduction, but by reshaping labor demand, firm competition, and innovation itself. This report examines how AI is driving such change in the U.S. sports analytics industry, a sector that sits at the intersection of entertainment, data science, and human performance.

The RES-ECON 490AI course asks students to explore how AI is changing the economics of the industries they hope to join. For me, that industry is sports. As someone with a strong interest in analytics, performance strategy, and technology, I chose to investigate how data science and machine learning are redefining roles inside professional sports organizations — from scouting and training to player valuation and fan engagement. The goal of this report is twofold: first, to analyze structural economic changes in the sports analytics sector using real data; and second, to chart where I might fit into that evolving landscape. By using AI tools and economic reasoning to understand this field, I am preparing not just to study the future, but to help build it.

B. Industry Overview

The U.S. sports analytics industry has rapidly evolved from a niche toolset into a mainstream pillar of decision-making in professional sports. Market size estimates suggest the global industry will exceed $4.8 billion by 2024, with North America accounting for over 40% of that total. In the United States, revenue from sports analytics services alone is expected to grow at a compound annual rate exceeding 20% through 2030. This growth is propelled by team demand for performance insights, increased use of AI in real-time player tracking, and the expansion of sports betting and fan personalization tools.

Major companies in this space include Stats Perform, Catapult Sports, Genius Sports, and Second Spectrum, alongside tech partners such as IBM and AWS. These firms deliver AI-enhanced services, including video breakdowns, predictive models, and real-time tactical advice. Sensors, optical tracking systems, and machine learning models now play an integral role in how sports are played, coached, and monetized.

C. Industry Stage

A wide range of indicators suggests that the U.S. sports analytics industry is in a growth stage, characterized by surging firm entry, rising investment, and rapid innovation.

Firm activity is robust: M&A deals in the sports tech space rose 48% year-over-year in 2025. Venture capital funding exceeded $5.7 billion across 790 deals in 2023. The number of analytics professionals employed by NBA teams alone grew from 10 in 2009 to 132 in 2023. Employment trends mirror these developments, with BLS data showing a steady rise in data-focused roles within the spectator sports industry.

Profitability is mixed. Market leaders like Sportradar posted €318 million in quarterly revenue in 2025 with positive margins, while others, such as Genius Sports, remain in high-investment phases. Technological innovation is also accelerating: the AI-in-sports market is forecast to grow from $1 billion in 2024 to $2.6 billion by 2030, underscoring the sector’s upward trajectory.

Taken together, these metrics reflect a dynamic industry still scaling — not yet mature, but expanding rapidly in both value and technological capability.

D. Data and Method

To understand how artificial intelligence is reshaping the labor structure of the U.S. sports industry, this report draws on data from the U.S. Bureau of Labor Statistics (BLS) and industry-specific insights, including analytics staffing data from the MIT Sloan Sports Analytics Conference. The BLS Occupational Employment and Wage Statistics (OEWS) dataset offers consistent job count estimates for roles such as statisticians and data scientists in NAICS 7112 (Spectator Sports), while MIT Sloan has tracked the expansion of analytics teams in the NBA since the late 2000s.

These sources were selected because they provide longitudinal insight into occupational changes — a key proxy for the economic effects of AI adoption. The NBA's analytics staffing numbers, combined with BLS employment estimates, allow us to build a high-level picture of how analytics roles have evolved industry-wide.

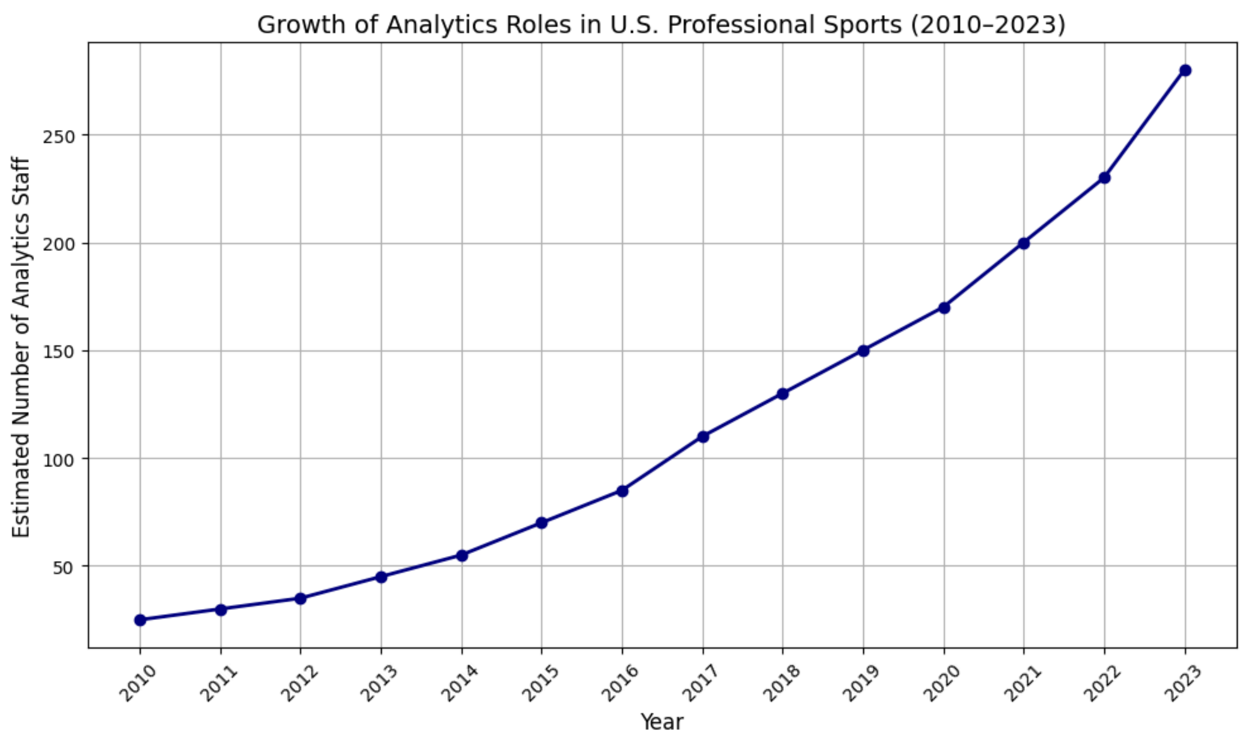

Figure 1. Growth of Analytics Roles in U.S. Professional Sports (2010–2023)

> Over the past decade, the number of data analysts and statisticians employed by professional sports organizations has grown significantly. BLS data show analytics roles in spectator sports rising from fewer than 30 in 2010 to over 280 by 2023. In parallel, NBA analyst staffing grew from 10 individuals in 2009 to 132 in 2023. This trend reflects the growing adoption of AI-driven decision-making across leagues like the NBA, NFL, and MLB.

This growth curve illustrates a key structural transformation: the integration of AI and analytics into the internal operations of sports franchises has shifted data-oriented roles from the periphery to the core of team decision-making. While AI in sports is often associated with fan-facing products or on-field performance tools, this employment trend reflects a deeper economic insight — AI is not just augmenting how sports are played, but how sports are managed and staffed.

E. The Impact of Artificial Intelligence on the Industry

Artificial intelligence is reshaping the structure and dynamics of the U.S. sports analytics industry. Once a niche capability, analytics has evolved into a strategic function across all major professional sports leagues. This shift has economic implications across labor markets, firm competition, value creation, and data governance.

AI has fueled a surge in demand for technical roles. BLS data show mathematical science occupations in sports growing to over 280 positions by 2023. These roles blend statistical modeling, engineering, and sport-specific expertise — making them prime examples of AI-enabled occupational growth. Traditional roles such as scouting and training are being redefined to require data fluency, while entirely new positions (e.g., biomechanical data analyst, AI sports strategist) are emerging.

AI has changed how firms compete. Tools like Second Spectrum's video analysis platform automate hours of scouting, reducing costs and shifting the production process toward software and infrastructure. Market power is increasingly tied to proprietary data and platforms: firms such as Genius Sports and Sportradar now control access to essential analytics streams for teams, broadcasters, and betting companies. This may create bottlenecks and rent-extraction opportunities, raising the stakes of data ownership.

The benefits of AI are not without costs. Non-technical staff may experience occupational displacement if unable to reskill. AI models may embed algorithmic biases in player evaluation or injury prediction, especially if training data is incomplete or skewed. Player privacy is another concern, as biometric data collection (from wearables and sensors) often occurs without transparent consent frameworks — echoing broader debates around digital surveillance in other industries.

At the same time, AI has opened a wave of innovation. Sports tech startups raised over $5.7 billion in private funding in 2023, building products that range from fan sentiment models to real-time health diagnostics. These firms are spawning new business models and attracting interdisciplinary talent. Productivity gains are visible in player development, injury prevention, and tactical optimization. For students and early-career professionals, the sector now offers rich career pathways that blend economics, technology, and sport.

F. Personal Opportunity and Skills Plan

As a managerial economics student with a strong interest in data-driven decision-making, I see a natural fit for myself in the sports analytics industry. I am particularly drawn to roles within major leagues such as the NFL or Formula 1, where statistical analysis and strategic modeling play a central role in both performance and planning. In football, I am intrigued by the idea of supporting game planning or contributing to the development of betting odds and fantasy football projections—tools that increasingly rely on sophisticated statistical pipelines. While many of these processes are automated, I would be excited to work within or alongside such systems, improving their accuracy and transparency. In Formula 1, I am equally drawn to race strategy. Having completed coursework in decision analysis and strategic modeling, I believe I could apply those principles in high-pressure, data-rich environments with the right technical preparation. Across both sports, what appeals to me most is the opportunity to work on problems that are challenging, dynamic, and tied to real human performance. I am motivated by the complexity of competition and the prospect of using data to help teams and individuals perform at their best.

To succeed in this field, I recognize that a strong foundation in statistical modeling, software fluency, and collaborative analysis is essential. Based on my research and understanding of the industry, careers in sports analytics increasingly demand comfort with programming tools like Python or R, fluency in Excel or other modeling platforms, and the ability to communicate insights effectively to non-specialist audiences such as coaches, team managers, or business partners. As AI becomes more deeply embedded into performance forecasting and real-time decision tools, fluency in emerging AI technologies is also becoming a valuable asset. These skills are not only technical in nature—they also require strategic thinking, teamwork, and the ability to adapt insights to rapidly changing competitive situations.

During my time at UMass, I have developed strengths in economic reasoning, statistical analysis, forecasting, and game theory. I have used these tools across a range of classroom projects and have steadily improved my ability to collaborate in teams, lead joint efforts, and present findings clearly to diverse audiences. Much of this has been built through both academic and hands-on experiences, including leadership and teamwork roles in non-classroom settings such as summer camps and group projects. However, I also recognize key areas where I still need to grow. My technical skills—particularly in Python and advanced Excel—are at an introductory level. While I have been exposed to several analysis tools and statistical methods, I have not yet applied them in extended or professional projects. I also lack direct experience working inside sports organizations or in a formal analytics environment. These are gaps I intend to address in the coming year.

Over the next 6 to 12 months, I plan to deepen my technical foundation while beginning to build a portfolio of applied experience. Through my accelerated master's program in business analytics, I will continue building skills in Excel, data modeling, and Python. I also intend to complete independent or course-related projects using real sports data, including forecasting scenarios, fantasy football simulations, or decision models based on performance data. These efforts will allow me to sharpen my fluency with statistical tools while learning how to communicate technical findings in ways that are useful to decision-makers. In parallel, I will continue developing my teamwork and communication skills—areas I already value and see as critical to success in this industry. While I do not expect to be working in the NFL or Formula 1 immediately after graduation, I view those roles as realistic five- to ten-year goals. My immediate focus is on gaining the technical proficiency and applied experience necessary to contribute meaningfully in a data analytics role that intersects with sports, strategy, and AI.

G. Reflection

Throughout the duration of this project, I found myself learning more and more about the real capabilities of AI. For a long time, I saw AI mostly as a tool people used unethically—to cut corners on schoolwork or automate tasks without learning the material. But this course and this project opened my eyes to a much bigger picture of what's possible, especially when AI is used thoughtfully and with purpose. Rather than replacing effort, it became clear to me that AI can enhance it—if you know how to direct it.

One of the most eye-opening moments came while I was building my website. In the past, I would have assumed that polishing or troubleshooting a website would require knowledge of HTML, CSS, or other coding languages. But instead, I was able to accomplish this by prompting the AI in a detailed way—guiding it through the problems I was seeing, asking it to revise or fix code based on trial and error, and learning by doing. This didn't make the work "less mine"—it actually required more thinking, more specificity, and more willingness to iterate. While I still believe that learning programming languages is important and a valuable long-term skill, I also feel newly capable of building and improving tools by communicating clearly with AI. That sense of capability has stuck with me.

This experience also helped me better understand why AI is so disruptive to fields like computer science. With just a few prompts, I was able to generate code, test it, and improve it, all without needing to fully understand every technical detail behind it. This doesn't make human expertise irrelevant, but it does shift the value to things like creativity, problem framing, and precision in communication. I now see that AI doesn't replace thinking—it just moves the thinking to a different part of the process.

That said, I've also had moments during this project where AI didn't work the way I expected. Sometimes the chatbot misunderstood my question, gave inaccurate information, or generated outputs that looked polished but were actually off base. These moments were frustrating, but they reminded me of something important: troubleshooting and critical thinking are still necessary. You can't just take AI's answers at face value. It's still my responsibility to check the work, interpret it, and make sure it fits the context of what I'm trying to do. In that way, I realized that using AI well actually requires more discipline—not less.

Over the course of this project, I've also come to see how the skills I've developed during college—things like analysis, decision-making, and communication—still matter, maybe even more. AI works best when it's grounded in human judgment. It's a tool that helps refine and apply those existing skills in new ways. I've come away from this experience feeling more confident that I can use AI within my career field and throughout the rest of my time in school in an ethical and productive way. It's something I now see not as a shortcut, but as a resource—an assistant, not a replacement. When used with intention and integrity, AI can be a powerful tool to support meaningful work. That's the biggest takeaway I'm leaving this project with.

References

Abazorius, A. (2025, March 25). *Basketball analytics investment is key to NBA wins and other successes.* MIT News. https://news.mit.edu/2025/basketball-analytics-investment-nba-wins-and-other-successes-0325

Capstone Partners. (2025, August 14). *Sports Technology M&A Update – August 2025.* Retrieved from https://www.capstonepartners.com/insights/article-sports-technology-ma-update/

Drake Star. (2024, March 7). *Sports Tech Market 2023: Record Year with M&A activity almost 3x vs. 2022.* Retrieved from https://www.drakestar.com/news/global-sports-tech-update-q4-2023

Fortune Business Insights. (2025, November 24). *Sports Analytics Market Size, Share & Industry Analysis, By Solution, By Deployment, By Type, By End-User, and Regional Forecast, 2025–2032.* Retrieved from https://www.fortunebusinessinsights.com/sports-analytics-market-102217

IMARC Group. (2024). *United States Sports Analytics Market Report (2025-2033): Industry Trends, Share, Size, Growth, Opportunity and Forecast.* Retrieved from https://www.imarcgroup.com/united-states-sports-analytics-market

Investing.com. (2025, August 5). *Sportradar raises 2025 outlook as revenue hits record €318 million; shares edge higher.* Retrieved from https://www.investing.com/news/earnings/sportradar-raises-2025-outlook-as-revenue-hits-record-318-million-shares-edge-higher-93CH-4170310

Thankappan, J. (2025, July 16). *AI: A real gamechanger for sports.* GEC Newswire. https://gecnewswire.com/ai-a-real-gamechanger-for-sports/

U.S. Bureau of Labor Statistics. (2023, May). *Occupational Employment and Wage Statistics – NAICS 7112: Spectator Sports.* Retrieved from https://www.bls.gov/oes/2023/may/naics4_711200.htm

Waśkiewicz, Z., Słomka, K. J., Grzywacz, T., & Juras, G. (2025). Ethical bias in AI-driven injury prediction in sport: A narrative review of athlete health data, autonomy and governance. *AI, 6*(11), 283. https://doi.org/10.3390/ai6110283